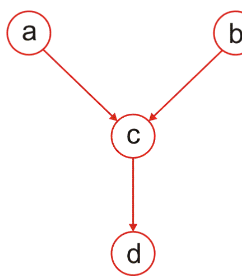
Task 1. Bayesian inference

Course: Bayesian Methods in Machine Learning 2023

$\mathbf{E}_{q(T)}[t_{nk}], E_{q(Z)}[z_n]$

Let us consider a model, which simulates students attending a university course. Let the audience of the course consist of students of a relevant faculty and students from other faculties. Denote the number of students of the relevant faculty with a and the number of students of all other faculties with b. The students of the relevant faculty attend the lectures with some probability p_1, while the rest of the students with the probability p_2. Let c denote the number of students attended a given lecture. Then, the random variable $c|a, b$ is a sum of two random variables, which are distributed according to $\text{Bin}(a, p_1)$ and $\text{Bin}(b, p_2)$, respectively. Now, the lecturer decides to check the attendance. Each student checks in for themselves and also, probably, checks in for their mate, who is not present at the lecture. The students check in for their mates with the probability p_3. Let d denote the number of students checked in at the lecture. Then, the random variable $d|c$ is a sum of c and a binomial random variable $\text{Bin}(c, p_3)$. It is only left to determine prior distributions for a and b to complete the probabilistic model. Let these variables be distributed uniformly within the bounds $[a_{min}, a_{max}]$ and $[b_{min}, b_{max}]$, respectively (discrete uniform distribution). Therefore, we have set up the following probabilistic model:

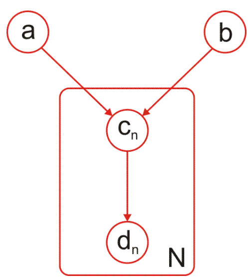


$$\begin{aligned}
&p(a, b, c, d) = p(d|c)p(c|a, b)p(a)p(b),\\
&d|c \sim c + \text{Bin}(c, p_3),\\
&c|a, b \sim \text{Bin}(a, p_1) + \text{Bin}(b, p_2),\\
&a \sim \text{Unif}[a_{min}, a_{max}],\\
&b \sim \text{Unif}[b_{min}, b_{max}].
\end{aligned} \tag{1}$$

We can consider a simplified version of the model 1. It is known that a binomial distribution $\text{Bin}(n, p)$ with a large number of trials n and small success probability p is well approximated by a Poisson distribution $\text{Poiss}(\lambda)$ with $\lambda = np$. Moreover, a sum of two Poisson distributions with parameters λ_1 and λ_2 is again a Poisson distribution with a parameter $\lambda_1 + \lambda_2$ (which does not hold for a sum of binomial random variables). Therefore, the following probabilistic model approximates the model 1:

$$\begin{aligned}
&p(a, b, c, d) = p(d|c)p(c|a, b)p(a)p(b),\\
&d|c \sim c + \text{Bin}(c, p_3),\\
&c|a, b \sim \text{Poiss}(ap_1 + bp_2),\\
&a \sim \text{Unif}[a_{min}, a_{max}],\\
&b \sim \text{Unif}[b_{min}, b_{max}].
\end{aligned} \tag{2}$$

Now, we can extend the model to attendance of several lectures. Let the attendance of different lectures be independent from each other. This gives us the model 3:



$$\begin{aligned}
&p(a, b, c_1, \dots, c_N, d_1, \dots, d_N) = p(a)p(b) \prod_{n=1}^{N} p(d_n|c_n)p(c_n|a, b),\\
&d_n|c_n \sim c_n + \text{Bin}(c_n, p_3),\\
&c_n|a, b \sim \text{Bin}(a, p_1) + \text{Bin}(b, p_2),\\
&a \sim \text{Unif}[a_{min}, a_{max}],\\
&b \sim \text{Unif}[b_{min}, b_{max}].
\end{aligned} \tag{3}$$

It is possible to simplify the model 3 similarly to the model 2:

$$
\begin{aligned}
&p(a, b, c_1, \ldots, c_N, d_1, \ldots, d_N) = p(a)p(b) \prod_{n=1}^{N} p(d_n|c_n)p(c_n|a, b), \\
&d_n|c_n \sim c_n + \mathrm{Bin}(c_n, p_3), \\
&c_n|a, b \sim \mathrm{Poiss}(ap_1 + bp_2), \\
&a \sim \mathrm{Unif}[a_{min}, a_{max}], \\
&b \sim \mathrm{Unif}[b_{min}, b_{max}].
\end{aligned} \tag{4}
$$

This task consists of three variants. The variant assignment scheme is attached to the task.

Variant 1

Consider the models 1 and 2 with the parameter values $a_{min} = 75$, $a_{max} = 90$, $b_{min} = 500$, $b_{max} = 600$, $p_1 = 0.1$, $p_2 = 0.01$, $p_3 = 0.3$.

1. Derive analytical formulae for all the required distributions.
2. Calculate expectation and variance for prior distributions $p(a)$, $p(b)$, $p(c)$, $p(d)$.
3. Observe how the forecast for the value c is refined as new indirect information arrives. To do this, plot graphs and find the expectation and variance for the distributions $p(c)$, $p(c|a)$, $p(c|b)$, $p(c|d)$, $p(c|a, b)$, $p(c|a, b, d)$ with parameters a, b, d equal to the expectation of their prior distributions, rounded to the nearest integer.
4. Determine which of the values a, b, d makes the greatest contribution to refining the forecast for the value c (in the sense of distribution variance). To do this, check whether it is true that $\mathrm{Var}[c|d] < \mathrm{Var}[c|b]$ and $\mathrm{Var}[c|d] < \mathrm{Var}[c|a]$ for any valid values a, b, d. Find the set of points (a, b) such that $\mathrm{Var}[c|b] < \mathrm{Var}[c|a]$. Are the sets $\{\, (a, b) \mid \mathrm{Var}[c|b] < \mathrm{Var}[c|a] \,\}$ and $\{\, (a, b) \mid \mathrm{Var}[c|b] \geq \mathrm{Var}[c|a] \,\}$ linearly separable? The answer must be justified!
5. Carry out time measurements to evaluate all necessary distributions $p(c)$, $p(c|a)$, $p(c|b)$, $p(c|d)$, $p(c|a, b)$, $p(c|a, b, d)$, $p(d)$.
6. Using the results of all the previous paragraphs, compare the two models. Show where the difference between them is most pronounced (give a specific example, not necessarily from the experiments above). Explain the reasons for this result.

Take the interval $[0, a_{max} + b_{max}]$ as the range of possible values for the random variable c, and the interval $[0, 2(a_{max} + b_{max})]$ for the random variable d.

The study must be conducted on a computer, however, additional points will be awarded for additional analytical derivations in paragraphs 2-4. When assessing the task, the efficiency of the program code will be taken into account - all the functions must work faster than a second on scalar inputs (for this, all the calculations have to be vectorized). For all paragraphs of the assignment, an analysis of the results must be carried out and conclusions must be drawn.

Variant 2

Consider the models 1 and 2 with the parameter values $a_{min} = 75$, $a_{max} = 90$, $b_{min} = 500$, $b_{max} = 600$, $p_1 = 0.1$, $p_2 = 0.01$, $p_3 = 0.3$.

1. Derive analytical formulae for all the required distributions.
2. Calculate expectation and variance for prior distributions $p(a)$, $p(b)$, $p(c)$, $p(d)$.
3. Observe how the forecast for the value b is refined as new indirect information arrives. To do this, plot graphs and find the expectation and variance for the distributions $p(b)$, $p(b|a)$, $p(b|d)$, $p(b|a, d)$ with parameters a, d equal to the expectation of their prior distributions, rounded to the nearest integer.
4. Determine the values of parameters p_1, p_2 for which the relative importance of parameters a, b for estimating the value of c changes. To do this, find the set of points $\{\, (p_1, p_2) \mid \mathrm{Var}[c|b] < \mathrm{Var}[c|a] \,\}$ with a, b equal to the expectation of their prior distributions rounded to the nearest integer. Are the sets $\{\, (p_1, p_2) \mid \mathrm{Var}[c|b] < \mathrm{Var}[c|a] \,\}$ and $\{\, (p_1, p_2) \mid \mathrm{Var}[c|b] \geq \mathrm{Var}[c|a] \,\}$ linearly separable? The answer must be justified!
5. Carry out time measurements to evaluate all necessary distributions $p(c)$, $p(c|a)$, $p(c|b)$, $p(b|a)$, $p(b|d)$, $p(b|a, d)$, $p(d)$.

6. Using the results of all the previous paragraphs, compare the two models. Show where the difference between them is most pronounced (give a specific example, not necessarily from the experiments above). Explain the reasons for this result.

Take the interval $[0, a_{max} + b_{max}]$ as the range of possible values for the random variable c, and the interval $[0, 2(a_{max} + b_{max})]$ for the random variable d.

The study must be conducted on a computer, however, additional points will be awarded for additional analytical derivations in paragraphs 2-4. When assessing the task, the efficiency of the program code will be taken into account - all the functions must work faster than a second on scalar inputs (for this, all the calculations have to be vectorized). For all paragraphs of the assignment, an analysis of the results must be carried out and conclusions must be drawn.

Variant 3

Consider the models 3 and 4 with the parameter values $a_{min} = 75$, $a_{max} = 90$, $b_{min} = 500$, $b_{max} = 600$, $p_1 = 0.1$, $p_2 = 0.01$, $p_3 = 0.3$, $N = 50$.

1. Derive analytical formulae for all the required distributions.
2. Calculate expectation and variance for prior distributions $p(a)$, $p(b)$, $p(c_n)$, $p(d_n)$.
3. Implement a sample generator $d_1, \ldots, d_N$ from the model for given values of parameters a, b.
4. Observe how the forecast for the value b is refined as new indirect information arrives. To do this, plot graphs and find the expectation and variance for the distributions $p(b)$, $p(b|d_1), \ldots, p(b|d_1, \ldots, d_N)$, where the sample is $d_1, \ldots, d_N$ 1) generated from the model with parameters a, b equal to the expectation of their prior distributions, rounded to the nearest integer and 2) $d_1 = \cdots = d_N$, where d_n is equal to the expectation of its prior distribution, rounded to the nearest integer. Carry out a similar experiment if the value of a is additionally known. Compare the results of two experiments.
5. Carry out time measurements to evaluate all necessary distributions $p(c_n)$, $p(d_n)$, $p(b|d_1, \ldots, d_n)$, $p(b|a, d_1, \ldots, d_n)$
6. Using the results of all the previous paragraphs, compare the two models. Show where the difference between them is most pronounced (give a specific example, not necessarily from the experiments above). Explain the reasons for this result.

Take the interval $[0, a_{max} + b_{max}]$ as the range of possible values for the random variable c, and the interval $[0, 2(a_{max} + b_{max})]$ for the random variable d.

The study must be conducted on a computer, however, additional points will be awarded for additional analytical derivations in paragraph 2. When assessing the task, the efficiency of the program code will be taken into account - all the functions must work faster than a second on scalar inputs a, b and one-dimensional input vectors c_n, d_n of length about 50 (for this, all the calculations have to be vectorized). For all paragraphs of the assignment, an analysis of the results must be carried out and conclusions must be drawn. Additional points may also be awarded for qualitative analysis in paragraph 4.